

Combinatorics – 1

Counting and bijections

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Theoretical minimum. Enumerative combinatorics has exactly three ways of proving that two quantities are equal: build a bijection, count one set in two ways, or sieve. Everything in this sheet is one of the three.

Definitions and basic counts. The number of ways to choose an ordered k -tuple of distinct elements of an n -set is $n(n-1)\cdots(n-k+1)$, of an unordered k -subset is $\binom{n}{k}$, and of a multiset of size k (“combinations with repetition”) is $\binom{n+k-1}{k}$ — the last by the *stars and bars* bijection with the arrangements of k stars and $n-1$ bars. The number of ways to split an n -set into blocks of prescribed sizes $k_1+\cdots+k_m=n$ is the multinomial coefficient $\frac{n!}{k_1!\cdots k_m!}$.

Fact 1 (a bijection is a proof). An identity between two counts is best proved by exhibiting a bijection between the sets they count: $\binom{n}{k} = \binom{n}{n-k}$ by complementation, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ by whether the last element is taken, $\sum_k \binom{n}{k} = 2^n$ by encoding a subset as a binary word. The bijection also transports extra information — statistics, weights, symmetries — which a computation does not.

Fact 2 (double counting). Counting the same set of pairs in two ways gives $\sum_v \deg v = 2|E|$, $k\binom{n}{k} = n\binom{n-1}{k-1}$ and the hockey-stick identity $\sum_{m \geq k} \binom{m}{k} = \binom{n+1}{k+1}$. The general form of the method, together with its non-combinatorial uses, is the subject of Methods of proof – 4.

Theorem 1 (inclusion–exclusion). For finite sets

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{\emptyset \neq I \subseteq [n]} (-1)^{|I|+1} \left| \bigcap_{i \in I} A_i \right|.$$

Proof: an element belonging to exactly $m \geq 1$ of the sets is counted $\sum_{j \geq 1} (-1)^{j+1} \binom{m}{j} =$

1 times, by the binomial theorem. The complementary form counts the elements avoiding all the A_i ; taking A_i to be the permutations fixing i gives the number of derangements

$$D_n = n! \sum_{j=0}^n \frac{(-1)^j}{j!}, \quad \frac{D_n}{n!} \rightarrow e^{-1},$$

and taking A_p to be the multiples of a prime divisor gives Euler's $\varphi(n) = n \prod_{p|n} (1 - 1/p)$.

Fact 3 (counting up to symmetry). When objects are counted up to the action of a group, the answer is an average of fixed-point counts (Burnside's lemma), proved and exercised in Abstract algebra – 4; it is not repeated here.

1. Prove the stars-and-bars formula, the multinomial theorem and the hockey-stick identity, each by an explicit bijection or a double count. Then prove that the number of monotone lattice paths from $(0, 0)$ to (a, b) is $\binom{a+b}{a}$, and count those that avoid a given point.
2. Prove the inclusion–exclusion formula and derive: the number of derangements and its asymptotics; the number of surjections from an n -set onto a k -set; and Euler's formula for $\varphi(n)$. Then compute the number of permutations of $[n]$ with no fixed point among the first k positions.
3. Let $A_n = \{1, 2, \dots, n\}$ with $n \geq 3$, and let F be the family of all non-constant functions $f: A_n \rightarrow A_n$ with $f(k) \leq f(k+1)$ and $f(k) = f(f(k+1))$ for $k = 1, \dots, n-1$. Find $|F|$. (IMC)
4. Let n be a positive integer. In how many ways can n be written as a sum $n = a_1 + a_2 + \dots + a_k$ of positive integers with $a_1 \leq a_2 \leq \dots \leq a_k \leq a_1 + 1$, the number k being arbitrary? (Putnam)
5. Let $\Omega = \{(x, y, z) \in \mathbb{Z}^3 : y + 1 \geq x \geq y \geq z \geq 0\}$. A frog moves along the points of Ω by jumps of length 1. For every positive integer n determine the number of paths from $(0, 0, 0)$ to (n, n, n) in exactly $3n$ jumps. (IMC)